Lab 10: File search based on metadata

What You Need for this lab

- Install Virtualbox : https://www.virtualbox.org/wiki/Downloads
- Install Kali 2021.4. : https://old.kali.org/kali-images/kali-2021.4/
    - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- A USB image

  https://www.dropbox.com/s/a0lhsmzeh68wk1i/NTFS.001

Step 1.

- Create a working folder

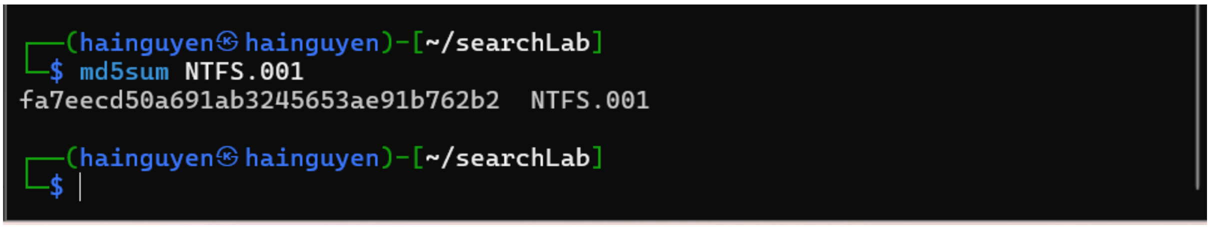
```
Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Jun 23 04:12:19 2025 from 192.168.1.7
┌──(hainguyen㉿hainguyen)-[~]
└─$ mkdir searchLab

┌──(hainguyen㉿hainguyen)-[~]
└─$ cd searchLab

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ 
```

ENG 10:21 AM 6/24/2025

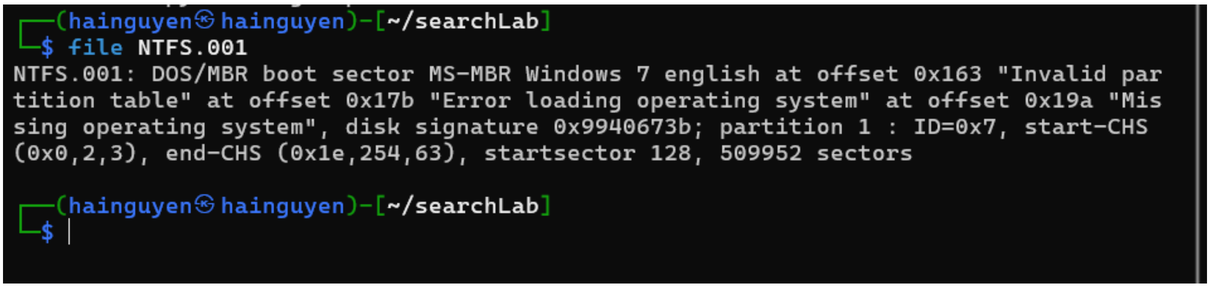
- Download a USB image

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ wget -q https://www.dropbox.com/s/a0lhsmzeh68wk1i/NTFS.001

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ ls -l
total 256544
-rw-rw-r-- 1 hainguyen hainguyen 262694912 Jun 23 23:25 NTFS.001

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ 
```

ENG 10:26 AM 6/24/2025

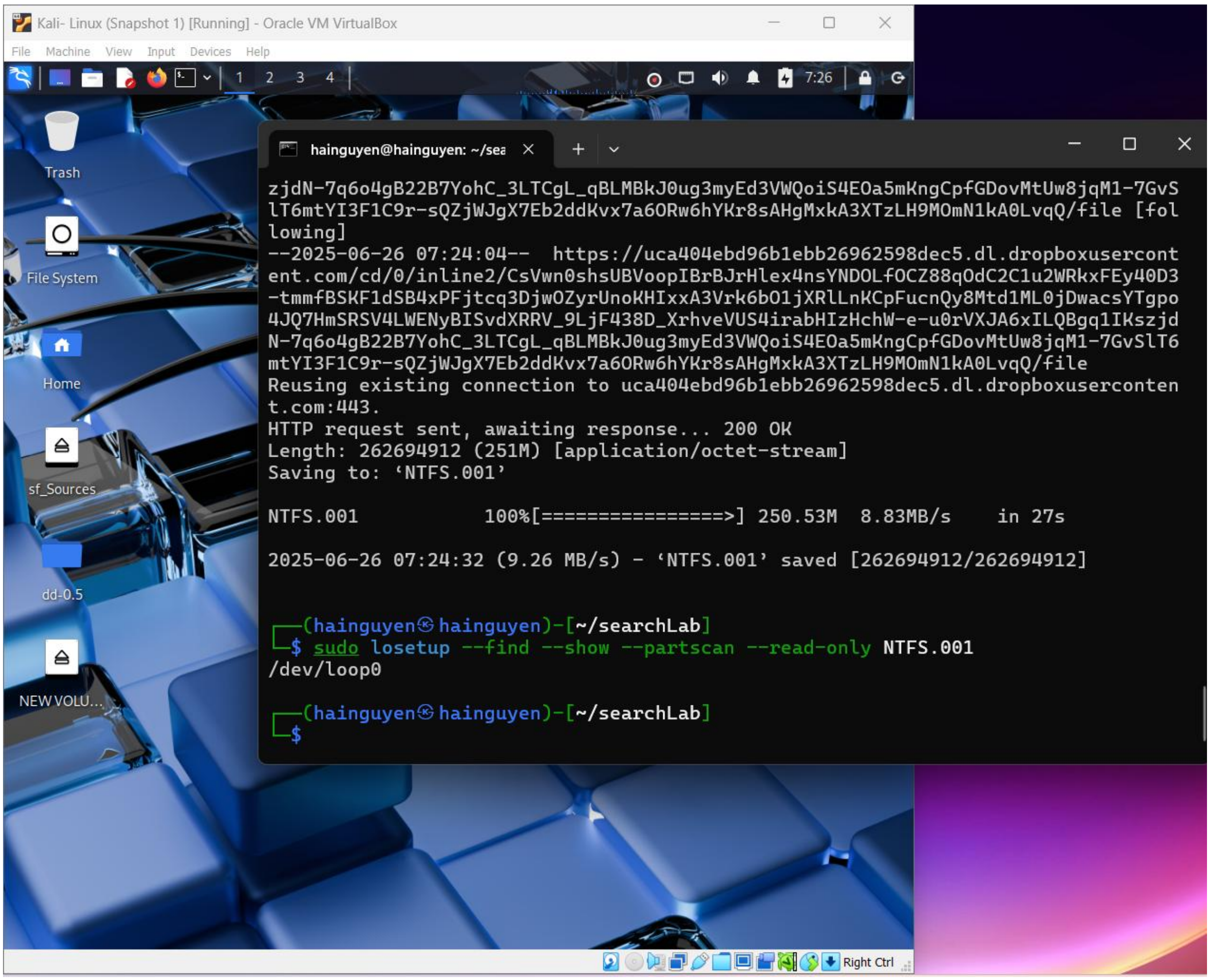


– Show image information



Step 2.

Mount the USB to Linux as a read-only loop device



– View file structure

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ tree /media/hainguyen/"NEW VOLUME"
/media/hainguyen/NEW VOLUME
├── forTeaching
│   ├── crack_word_lab.TXT
│   ├── encrypted_file_123abc_2013_v.docx
│   ├── hash.txt
│   ├── HelloFAT.docx
│   ├── how_to_crack_pwd_123.pdf
│   ├── how_to_crack_pwd_abc123.pdf
│   ├── M57-Jean_Solution.pdf
│   ├── nps-2008-jean_outlook.pst
│   ├── office2john.py
│   ├── pdf2john.py
│   └── pdfM57-Jean-hash.txt
├── HelloNTFS.docx
└── System Volume Information
    ├── IndexerVolumeGuid
    └── WPSettings.dat

3 directories, 14 files

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$
```

– Syntax

$ find [where to start searching from] [expression determines what to find]

```
hainguyen@hainguyen: ~/sea

NAME
       find - search for files in a directory hierarchy

SYNOPSIS
       find  [-H]  [-L]  [-P]  [-D debugopts] [-Olevel] [starting-point...]
       [expression]

DESCRIPTION
       This manual page documents  the  GNU  version  of  find.   GNU  find
       searches  the  directory tree rooted at each given starting-point by
       evaluating the given expression from left to right, according to the
       rules of precedence (see section OPERATORS), until  the  outcome  is
       known (the left hand side is false for and operations, true for or),
       at which point find moves on to the next file name.  If no starting-
       point is specified, `.' is assumed.

       If  you are using find in an environment where security is important
       (for example if you are using it  to  search  directories  that  are
       writable  by  other users), you should read the `Security Considera-
       tions' chapter of the findutils documentation, which is called Find-
       ing Files and comes with findutils.  That document also  includes  a
       lot  more  detail  and  discussion than this manual page, so you may
       find it a more useful source of information.

 Manual page find(1) line 2 (press h for help or q to quit)
```

– Find Files/directories starts from Current Directory (recursively)

```
hainguyen@hainguyen: /med

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ cd /media/hainguyen/"NEW VOLUME/"

┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find .
.
./forTeaching
./forTeaching/crack_word_lab.TXT
./forTeaching/encrypted_file_123abc_2013_v.docx
./forTeaching/hash.txt
./forTeaching/HelloFAT.docx
./forTeaching/how_to_crack_pwd_123.pdf
./forTeaching/how_to_crack_pwd_abc123.pdf
./forTeaching/M57-Jean_Solution.pdf
./forTeaching/nps-2008-jean_outlook.pst
./forTeaching/office2john.py
./forTeaching/pdf2john.py
./forTeaching/pdfM57-Jean-hash.txt
./HelloNTFS.docx
./System Volume Information
./System Volume Information/IndexerVolumeGuid
./System Volume Information/WPSettings.dat

┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ 
```

-name and *-iname*: find file/directory names

- find a file

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find ./forTeaching -name hash.txt
./forTeaching/hash.txt

┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -name hash.txt
./forTeaching/hash.txt

┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ 
```

- find a file and ignore case

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -name hash.TXT

┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -iname hash.TXT
./forTeaching/hash.txt

┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ 
```

- find a directory

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -name forTeaching
./forTeaching
```

type: specify string types

- Specify the search type *d: directory*

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -type d -name forTeaching
./forTeaching
```

- Specify the search type *f: file*

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -type f -name hash.txt
./forTeaching/hash.txt
```

- Search all files with extension *.txt*

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -type f -name "*.txt"
./forTeaching/hash.txt
./forTeaching/pdfM57-Jean-hash.txt
```

- Match all files which first and four letters are "*h*"

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -type f -name "h??h*"
./forTeaching/hash.txt
```

- Match all files contains "*123*" (in the middle of the file name)

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find . -type f -name "*123*"
./forTeaching/encrypted_file_123abc_2013_v.docx
./forTeaching/how_to_crack_pwd_123.pdf
./forTeaching/how_to_crack_pwd_abc123.pdf
```

- Find Files With 777 Permissions

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ find ./forTeaching -type f -perm 777 -print
./forTeaching
./forTeaching/crack_word_lab.TXT
./forTeaching/encrypted_file_123abc_2013_v.docx
./forTeaching/hash.txt
./forTeaching/HelloFAT.docx
./forTeaching/how_to_crack_pwd_123.pdf
./forTeaching/how_to_crack_pwd_abc123.pdf
./forTeaching/M57-Jean_Solution.pdf
./forTeaching/nps-2008-jean_outlook.pst
./forTeaching/office2john.py
./forTeaching/pdf2john.py
./forTeaching/pdfM57-Jean-hash.txt

┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$
```

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ ls -l ./forTeaching
total 4080
-rwxr-xr-x 1 hainguyen hainguyen      57 Oct  2  2018 crack_word_lab.TXT
-rwxr-xr-x 1 hainguyen hainguyen   17920 Oct  2  2018 encrypted_file_123abc_2013_v.docx
-rwxr-xr-x 1 hainguyen hainguyen     194 Oct  2  2018 hash.txt
-rwxr-xr-x 1 hainguyen hainguyen   11514 Oct  8  2018 HelloFAT.docx
-rwxr-xr-x 1 hainguyen hainguyen 1185208 Oct  1  2018 how_to_crack_pwd_123.pdf
-rwxr-xr-x 1 hainguyen hainguyen  380207 Oct  1  2018 how_to_crack_pwd_abc123.pdf
-rwxr-xr-x 1 hainguyen hainguyen   96290 Oct  5  2018 M57-Jean_Solution.pdf
-rwxr-xr-x 1 hainguyen hainguyen 2326528 Jul 20  2008 nps-2008-jean_outlook.pst
-rwxr-xr-x 1 hainguyen hainguyen  134772 Oct  2  2018 office2john.py
-rwxr-xr-x 1 hainguyen hainguyen   13904 Oct  5  2018 pdf2john.py
-rwxr-xr-x 1 hainguyen hainguyen     191 Oct  5  2018 pdfM57-Jean-hash.txt
```

Step 3.

- **“atime”, “mtime” and “ctime”**

```
┌──(hainguyen㉿hainguyen)-[/media/hainguyen/NEW VOLUME]
└─$ cd ~/searchLab

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ echo hello world > myFile.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ ls -l
total 256548
-rw-rw-r-- 1 hainguyen hainguyen        12 Jun 26 08:58 myFile.txt
-rw-rw-r-- 1 hainguyen hainguyen 262694912 Jun 26 08:44 NTFS.001

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ echo hello world again! >> myFile.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ ls -l
total 256548
-rw-rw-r-- 1 hainguyen hainguyen        31 Jun 26 08:59 myFile.txt
-rw-rw-r-- 1 hainguyen hainguyen 262694912 Jun 26 08:44 NTFS.001

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ cat myFile.txt
hello world
hello world again!

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$
```

Can you recreate the file changing history? Answer and explaine

```
hainguyen@hainguyen: ~/sea

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ stat myFile.txt
  File: myFile.txt
  Size: 31              Blocks: 8          IO Block: 4096   regular file
Device: 8,1     Inode: 2752797     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/hainguyen)   Gid: ( 1000/hainguyen)
Access: 2025-06-26 08:59:27.142202347 -0400
Modify: 2025-06-26 08:59:09.837554812 -0400
Change: 2025-06-26 08:59:09.837554812 -0400
 Birth: 2025-06-26 08:58:20.400850156 -0400

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$
```

Answer

- Timestamp changes when changing permission

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ rm testTimeStamp.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ echo test for deletion > testTimeStamp.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ chmod 644 testTimeStamp.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ ls -l
total 256552
-rw-r--r-- 1 hainguyen hainguyen        31 Jun 26 08:59 myFile.txt
-rw-r--r-- 1 hainguyen hainguyen 262694912 Jun 26 08:44 NTFS.001
-rw-r--r-- 1 hainguyen hainguyen        18 Jun 26 09:05 testTimeStamp.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ sudo chmod 664 testTimeStamp.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ ls -l
total 256552
-rw-r--r-- 1 hainguyen hainguyen        31 Jun 26 08:59 myFile.txt
-rw-r--r-- 1 hainguyen hainguyen 262694912 Jun 26 08:44 NTFS.001
-rw-rw-r-- 1 hainguyen hainguyen        18 Jun 26 09:05 testTimeStamp.txt

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$
```

Find Files With "*–perm=664*" Permissions ?

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ ls -l
total 256552
-rw-r--r-- 1 hainguyen hainguyen        31 Jun 26 08:59 myFile.txt
-rw-r--r-- 1 hainguyen hainguyen 262694912 Jun 26 08:44 NTFS.001
-rw-rw-r-- 1 hainguyen hainguyen        18 Jun 26 09:05 testTimeStamp.txt
```

Answer

Find Files With */u=r /u=w /u=x* Permissions

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ ls -l
total 256552
-rw-r--r-- 1 hainguyen hainguyen        31 Jun 26 08:59 myFile.txt
-rw-r--r-- 1 hainguyen hainguyen 262694912 Jun 26 08:44 NTFS.001
-rw-rw-r-- 1 hainguyen hainguyen        18 Jun 26 09:05 testTimeStamp.txt
```

Answer

Find Changed Files in Last 1 Hour ?

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ find . -type f -cmin -60
./NTFS.001
./myFile.txt
./testTimeStamp.txt
```

Answer

Find Accessed Files in Last 1 Hour ?

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ date
Thu Jun 26 09:08:57 AM EDT 2025

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ find . -type f -amin -60
./NTFS.001
./myFile.txt
./testTimeStamp.txt
```

Answer

Find files that size that is great than 10MB ?

```
┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ find . -size +10M
./NTFS.001

┌──(hainguyen㉿hainguyen)-[~/searchLab]
└─$ 
```

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 10.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 10 From YOUR NAME**", replacing "YOUR NAME" with your real name.